Session 07 – Integrating React with APIs (Assignment)

1. Objective

Session 07 focused on integrating the React frontend with the FastAPI backend API.

Main objectives:

- Connect React to FastAPI using Axios.
- Create a reusable Axios client.
- Organize API calls into separate modules.
- Implement shared error handling.
- Retrieve product list and product details from the backend API.

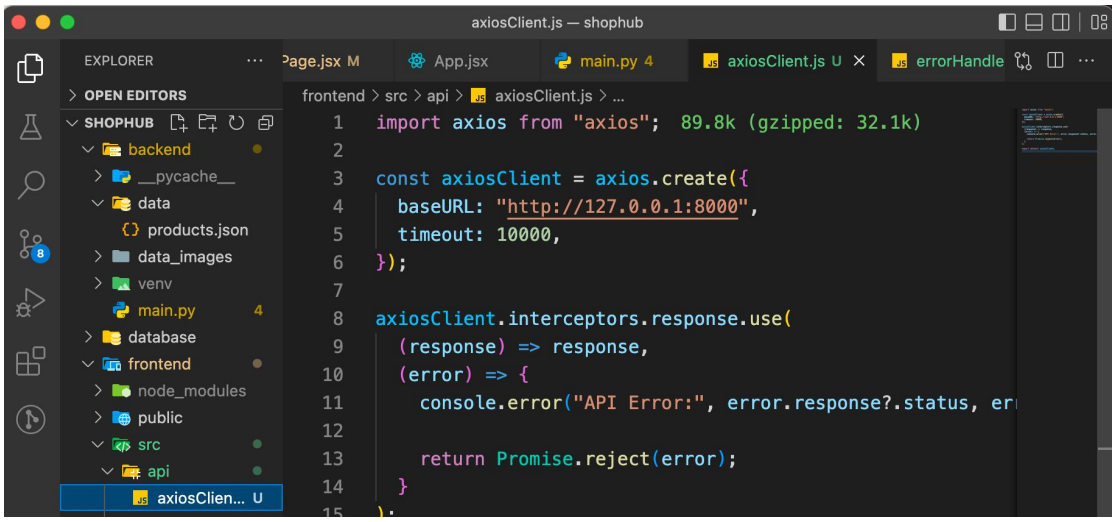
2. Main Concepts

2.1 Axios

Axios is an HTTP client used to communicate with backend APIs.

Features used:

- axios.create()
- GET requests
- async/await



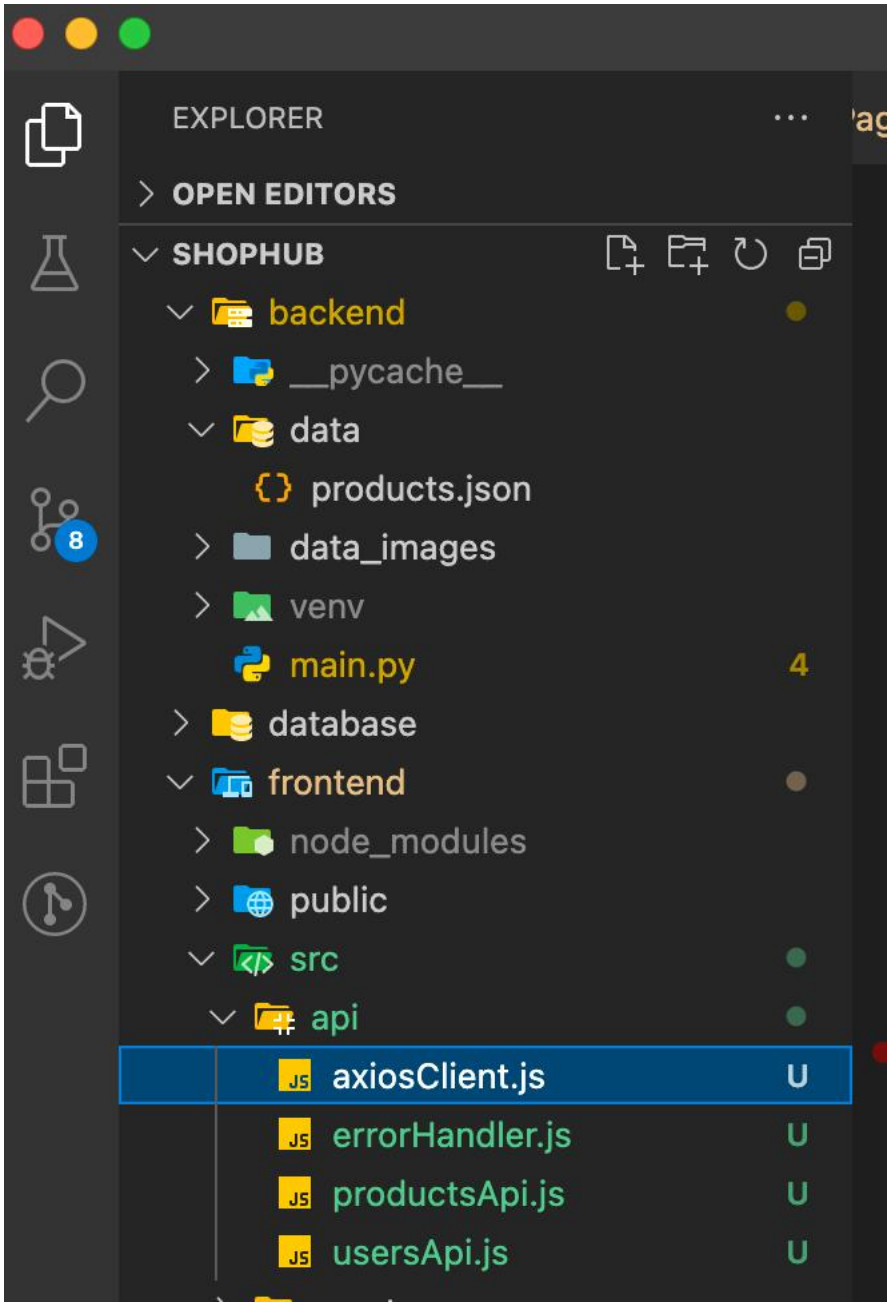
2.2 API Modules

API logic was separated into reusable modules:

- productsApi.js
- usersApi.js

Benefits:

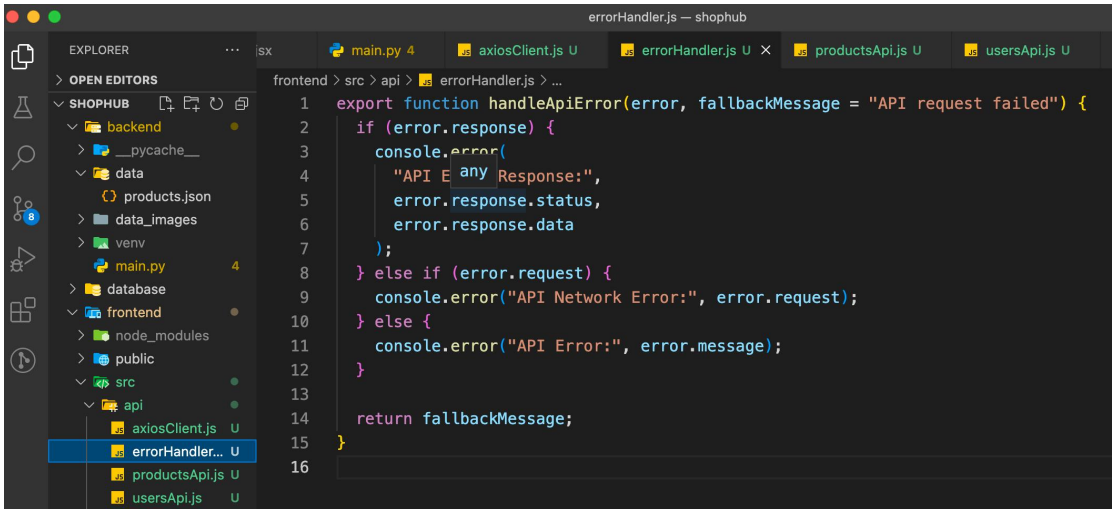
- Cleaner code
- Easier maintenance
- Reusable API functions



2.3 Error Handling

A shared function handleApiError() was created to handle:

- Server errors (4xx, 5xx)
- Network errors
- Other unexpected errors



3. Implementation

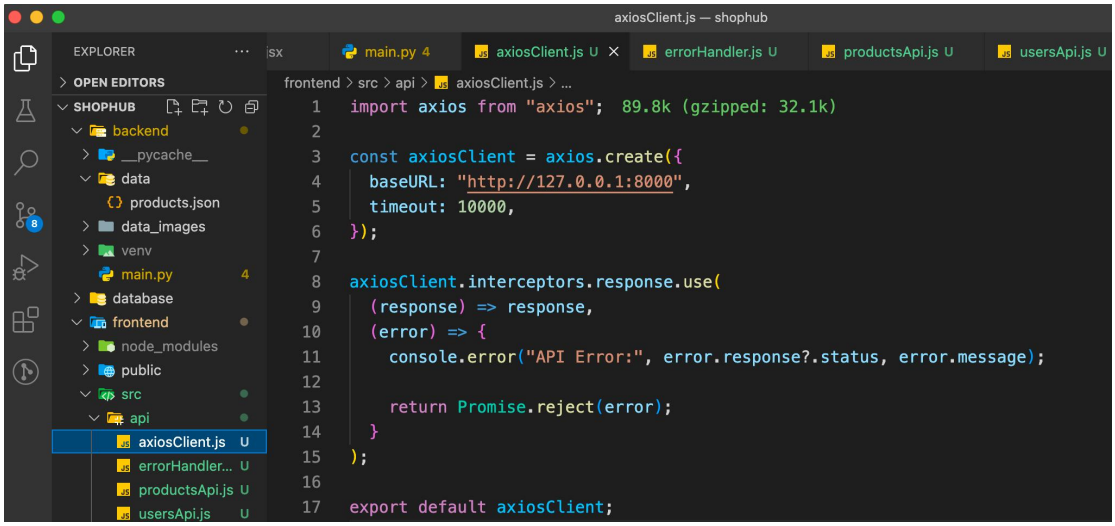
3.1 Axios Client

Created:

src/api/axiosClient.js

Configured:

- baseURL
- timeout
- response interceptor



3.2 Products API

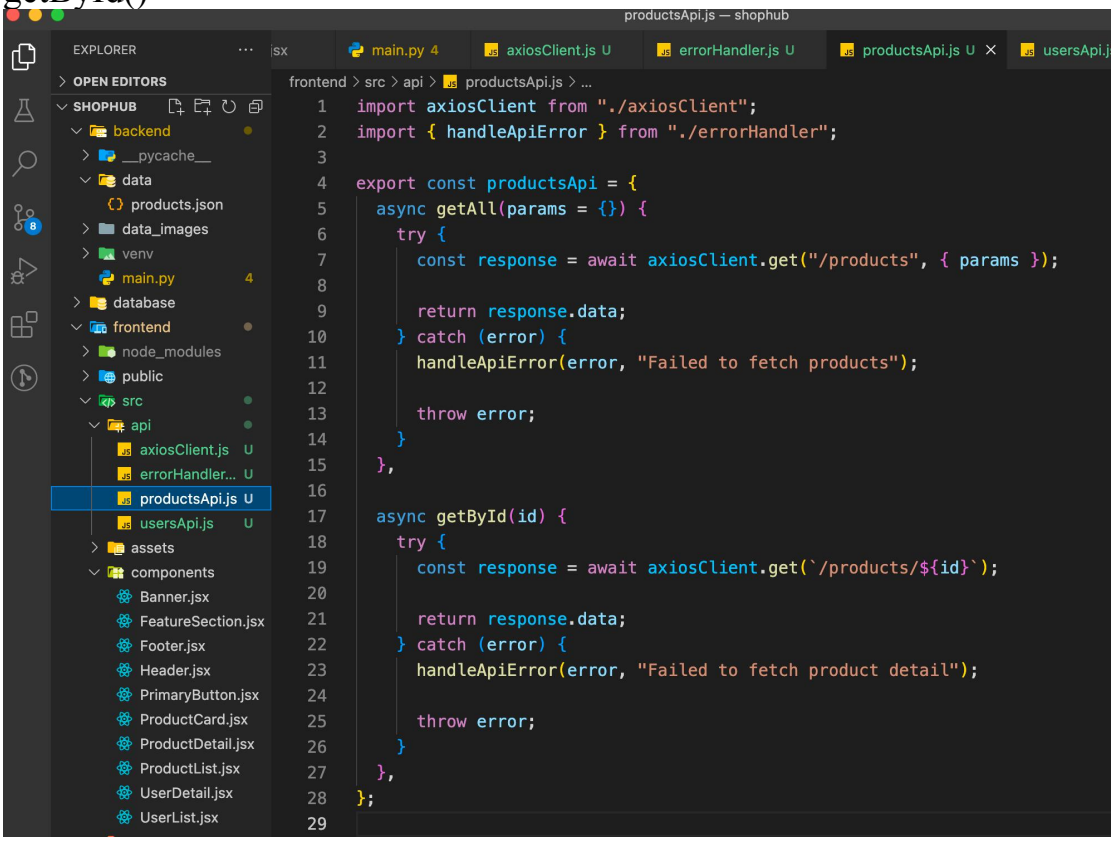
Created:

src/api/productsApi.js

Functions:

- getAll()

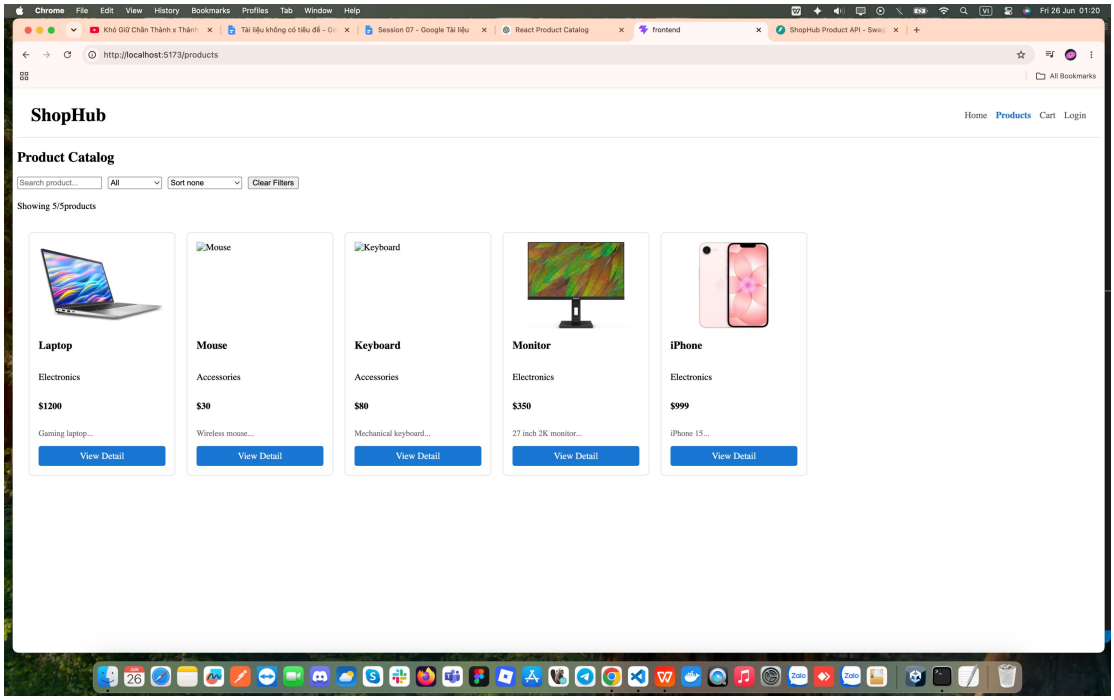
- getById()



3.4 Product List Integration

Updated ProductPage:

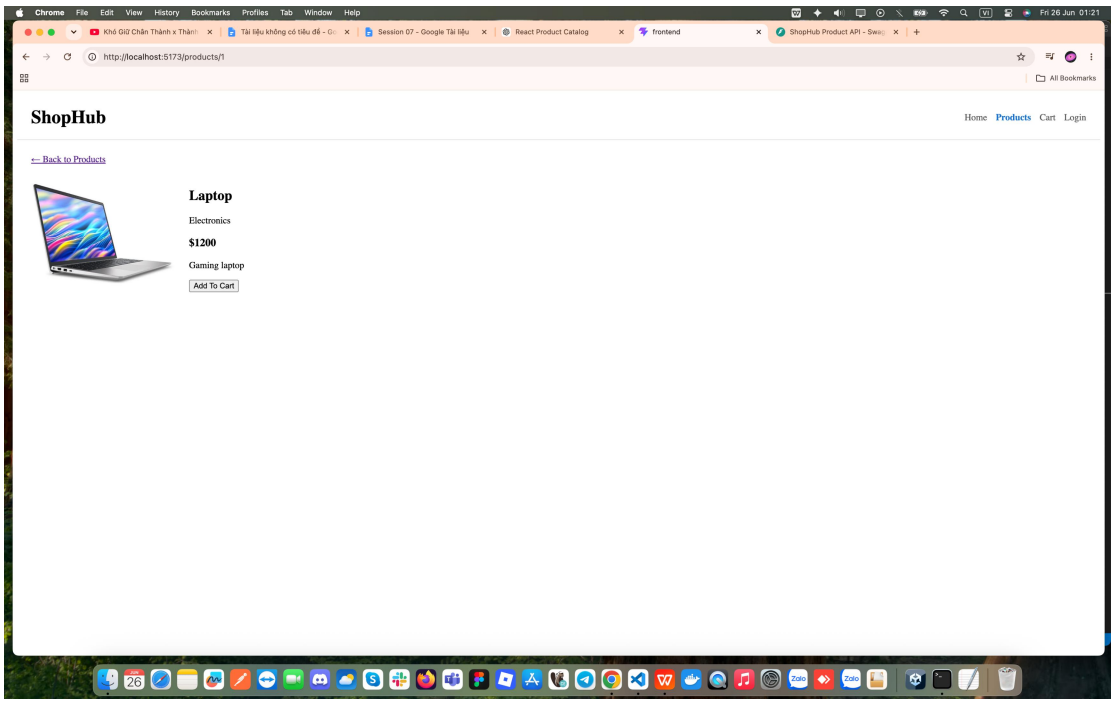
- Uses productsApi.getAll()
- Loads data from FastAPI backend
- Displays product list dynamically



3.5 Product Detail Integration

Updated ProductDetailPage:

- Uses productsApi.getById(id)
- Retrieves product detail from backend
- Displays product information and image

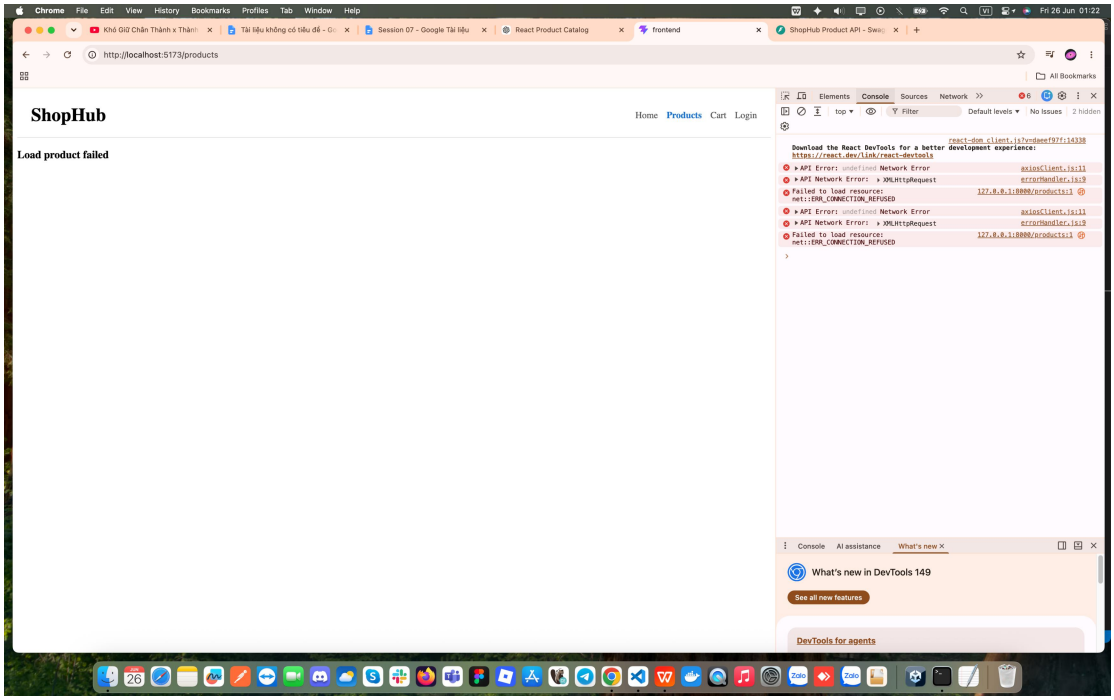


3.6 Error Handling Test

Tested by stopping the backend server.

Result:

- Error logged in browser console.
- Friendly message displayed on the UI.



4. Results

After completing Session 07:

- React successfully communicates with FastAPI.
- Product data is loaded from backend APIs.
- Product detail pages use dynamic API requests.
- API logic is centralized through reusable modules.
- Error handling is standardized across the application.
- Frontend architecture is more scalable and maintainable.